Exploring The Visual Feature Space for Multimodal Neural Decoding

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Abstract

The intrication of brain signals drives research that leverages multimodal AI to align brain modalities with visual and textual data for explainable descriptions. However, most existing studies are limited to coarse interpretations, lacking essential details on object descriptions, locations, attributes, and their relationships. This leads to imprecise and ambiguous reconstructions when using such cues for visual decoding. To address this, we analyze different choices of vision feature spaces from pre-trained visual components within Multimodal Large Language Models (MLLMs) and introduce a zero-shot multimodal brain decoding method that interacts with these models to decode across multiple levels of granularities. To assess a model's ability to decode fine details from brain signals, we propose the Multi-Granularity Brain Detail Understanding Benchmark (MG-BrainDub). This benchmark includes two key tasks: detailed descriptions and salient question-answering, with metrics highlighting key visual elements like objects, attributes, and relationships. Our approach enhances neural decoding precision and supports more accurate neuro-decoding applications. Code is available at https://github.com/weihaox/VINDEX.

1. Introduction

Human brain signals are inherently complex and not directly interpretable, posing a significant challenge for decoding brain activity. To address this, recent studies have focused on translating brain signals associated visual stimuli into multimodal explanations, such as textual descriptions [13, 42], depth information [54], bounding boxes [53], or direct image reconstruction [35, 41, 42]. Although these efforts have made notable strides, achieving a fine-grained understanding of brain modalities remains an unresolved challenge.

One line of research [42, 49, 53] has attempted to translate brain activations from visual stimuli into textual captions. However, existing methods typically generate one-sentence descriptions with limited details, failing to capture the full spectrum of visual perception. The most straightforward approach [43] is to use off-the-shelf MLLMs [25, 29] to generate detailed descriptions from paired images for each brain recording, then train a model to decode these pseudo-annotations from brain signals. However, this practice is computationally inefficient and may introduce bias or errors, making it impractical for generalizable brain decoding. The other key issue is the evaluation of brain-decoded detailed captions, which can be hundreds of words long, rendering conventional metrics unsuitable. Specifically, traditional captioning metrics [4, 36, 51] which mostly rely on n-gram matching between candidate and reference captions, are highly sensitive to stylistic variations, leading to inconsistent evaluations. Model-based metrics either use outdated text encoders [3] or struggle with long text due to token length limitations [15, 38]. While concurrent studies [43] attempt to generate detailed captions for visual decoding, they lack reliable evaluations of caption quality and accuracy.

In this paper, we address the two critical issues. For the method, we investigate the exploitation of various feature spaces from the visual components within MLLMs to enable fine-grained multimodal explanations from neural activations. We present VINDEX (VIsual experts for multimodal Neural DEcoding eXploration), which leverages diverse feature spaces for multimodal neural decoding. We further introduce a denoiser-based alignment strategy that efficiently maps brain tokens to image tokens, allowing for integration into pre-trained MLLMs. This approach facilitates decoding across multiple levels of granularities without the need for additional textual or spatial annotations. For evaluation, we propose the Multi-Granularity Brain Detail Understanding Benchmark (MG-BrainDub) to discern a model's ability to decode fine-grained details from brain signals. MG-BrainDub consists of two key tasks: detailed descriptions and salient question-answering, with metrics focusing on key visual elements, including objects, attributes, and relationships, derived from both candidate and ground truth captions. Our contributions are summarized as follows:

- Through systematic analysis, we exploit representative feature spaces of visual components within MLLMs for brain signal alignment and multimodal brain interaction.
- We introduce a zero-shot multimodal brain decoding method for multi-granular decoding without the need for additional textual or spatial annotations during training.
- We construct a benchmark with metrics for fine-grained

brain interpretation, evaluating a model's ability to decode detailed descriptions and perform complex reasoning.

2. Related Work

Multimodal Brain Perception. Recent advancements in decoding multimodal cues from brain signals, facilitated by MLLMs, have achieved unprecedented performance [35, 41, 49, 54]. These methods typically map brain responses, captured through functional magnetic resonance imaging (fMRI), to more commonly used modalities that can be processed by pre-trained models for subsequent information extraction [20, 40, 56]. For instance, Takagi and Nishimoto [49] connect fMRI data with CLIP text embeddings and the latent space of Stable Diffusion (SD) [40] using ridge regression to reconstruct visual stimuli. Xia *et al*. [54] extract semantic, depth, and color features from fMRI data to reconstruct images using a depth- and color-conditioned SD model. Several methods [13, 42, 49, 54] aim to create readable multimodal explanations of visual stimuli, such as textual description or depth estimation, but often rely on corresponding external supervision to guide the multimodal training. In contrast, UMBRAE [53] has recently demonstrated the ability to directly decode textual descriptions and bounding boxes from fMRI data using only image-fMRI pairs during training, without requiring additional annotations. This method aligns image features with visual encoders and integrates them into MLLMs. Building on this foundation, our proposed approach aims to further elucidate feature space representations, extracting more granular information while introducing metrics for evaluation.

Multimodal Large Language Models. Expanding Large Language Models (LLMs) to incorporate additional modalities, such as images, has recently attracted significant interest [10, 29, 50, 57]. These models typically comprise three components: a frozen image encoder, a trainable connector, and a frozen or finetuned LLM. The connector bridges the gap from the image feature space to the LLM token space, which can be implemented as a linear layer [7], a multilayer perceptron (MLP) [29], or a lightweight transformer [17]. Recent studies have shown that using pre-trained CLIP [38] as the visual component can lead to compromised performance in visual tasks, particularly in grounding capabilities [7, 50]. To overcome this limitation, researchers have explored ways to more effectively utilize the visual component to enhance perception across diverse scenarios. This includes employing a mixture of vision encoders [19, 44, 50, 59] pre-trained on various tasks as expert models and investigating the feature spaces [6, 57] within CLIP. Drawing inspiration from these efforts, we exploit brain alignment with feature spaces in MLLM's visual component for multi-granular representation to facilitate zero-shot multimodal brain decoding.

Multimodal-Brain Alignment. The prevailing practice for brain alignment is to map neural modalities into a shared multimodal space [13, 35, 41, 53], which can be categorized into generative alignment and discriminative alignment. The first category, exemplified by OneLLM [13], employs generative training to learn multimodal alignment, connecting multimodal inputs, including brain signals, with an LLM. Methods in the second category align brain signals within a pre-trained embedding space, such as CLIP [38], using techniques like linear regression [35, 49], contrastive learning [54], diffusion priors [41], or feature reconstruction [53]. MEVOX [55] ensembles multiple task-specific experts to capture distinct aspects of brain perception. The brain encoder learns to align with omni-contextual explanations. Most of these alignment strategies require per-subject training or subject-specific annotations, which can lead to scalability issues in practice. While UMBRAE [53] performs zero-shot multimodal brain decoding, its regression-based alignment is deterministic. We analyze the limitations of this strategy and propose incorporating a novel denoising optimization objective to enhance the brain-feature alignment.

3. Method

Current MLLMs [7, 29] comprise three components: a pre-trained vision encoder, a pre-trained LLM, and a connector that bridges vision and language models. Given significant discrepancies in data scale and challenges associated with acquiring large-scale brain datasets, we argue that aligning brain signals with pre-trained visual components, rather than training from scratch [13, 43], is a more effective and efficient approach. This not only alleviates the need to construct extensive text for training MLLMs but also enables zero-shot capability in decoding multimodal cues, allowing for versatile usage across different MLLMs. Thus, we present the rationale behind our method design in three key aspects: (1) the selection of vision feature spaces (Sec. 3.1), (2) the alignment strategy to connect activations with pre-trained MLLMs (Sec. 3.2), and (3) the development of a benchmark necessary for fine-grained brain perception (Sec. 4).

3.1. Exploring Feature Spaces

We explore four distinct feature spaces, drawing inspiration from prior MLLM research that leverage different feature representations. By aligning brain signals across these feature spaces, we aim to capture multi-granular representations in MLLMs, facilitating zero-shot multimodal brain decoding. The four feature space choices are illustrated in Fig. 1.

Single Encoder. This represents using features from a single pre-trained vision encoder (*e.g.* CLIP [38]) for brain alignment, which is the most commonly used in MMLMs.

Mixture of Encoders. Encoders trained on diverse tasks allow us to explore the distinct advantages of different vision experts. These include contrastive image-text learning

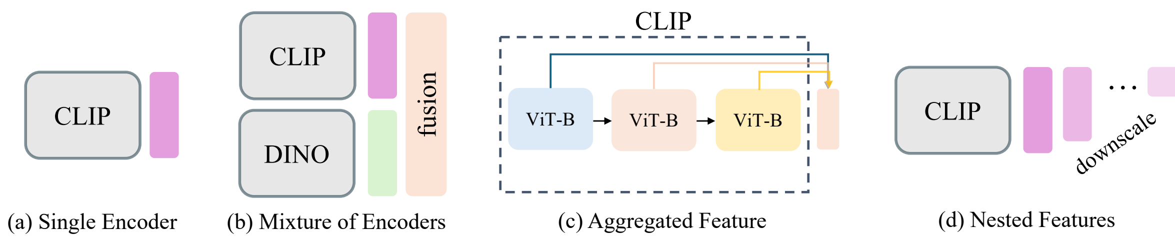

Figure 1. **Feature Spaces.** We aim to capture multi-granular representations for zero-shot multimodal brain decoding by aligning brain signals across different feature spaces in MLLM's visual component. The explored representative feature spaces include: (a) Single Encoder, which utilizes a single encoder to extract selected features; (b) Mixture of Encoders, using hybrid featured from different vision experts specialized in task-specific domains; (c) Aggregated Feature, fusing features from different layers within the same image encoder; and (d) Nested Features, downscaling visual representations in a hierarchical coarse-to-fine structure with different perceptual granularities.

CLIP [38], self-supervised learning DINO [34], and object-centric models pre-trained for various tasks such as segmentation (SAM [22]), detection (EVA [11]), and recognition (Pix2Struct [24]). These task-specific vision encoders allow MLLMs to achieve optimal performance within respective pre-training domains [19, 45]. Aligning brain signals with features from these models allows for the capture of distinct facets of visual perception. Here, we explore the integrated features from CLIP [38] and DINO [34] encoders for simplicity while demonstrating the core concept of aligning brain signals with a mixture of vision experts. CLIP effectively encodes semantic information, while DINO excels in capturing spatial details such as object location and boundaries.

Aggregated Feature. Existing MLLMs typically use features from a selected layer in the visual component as the visual representation input for LLMs. Features from different levels—such as shallow, middle, and deep layers—are recognized for capturing different characteristics of the image [57]. Therefore, by encapsulating information from multiple layers, we can enrich the visual input for the LLM. Specifically, we partition features from selected layers into several groups. Features within each group from adjacent layers are integrated along the channel dimension to reduce redundancy and mitigate high dimensionality. The features from each group are then concatenated with the features from the final layer before being fed into the connector.

Nested Features. MLLMs typically project feature images from the vision encoder into a fixed-length set of visual tokens [29]. Longer token sequences generally yield better results but require more resources. Recent studies have found that the optimal token length is task-specific, laying the foundation for visual token pruning and merging to dynamically adjust the sequence length [5, 21]. Therefore, we downscale selected features using a set of downsampling factors and produce nested features. This approach offers a flexible and controllable method to encode visual content as nested sets of visual tokens with a hierarchical structure, from coarse to fine details—much like a Matryoshka doll [6, 23]. The nested features capture structured information across multiple coarse-to-fine levels, enabling adaptable token lengths that align with different perceptual granularities.

3.2. Denoising for Multimodal Brain Alignment

This section outlines training a brain encoder using features from Sec. 3.1 as ground truth. Given brain responses $s \in \mathbb{R}^{1 \times L_s}$ (with L_s indicating the dimension of input brain data) and associated visual stimuli $v \in \mathbb{R}^{W \times H \times C}$ (representing image width, height, and channels), we train the brain encoder $\mathcal{B}$ to minimize the distance between brain features $\mathbf{b}$ and target image features $\mathbf{v}$ from the vision encoder $\mathcal{V}$, aiming for a close approximation $\mathcal{B}(s) \approx \mathcal{V}(v)$. Toward the objective, the most straightforward alignment is through element-wise feature reconstruction [53]:

$$\mathcal{L}_{\mathrm{R}} = \mathbb{E}_{\mathbf{b} \sim \mathbf{B}, \mathbf{v} \sim \mathbf{V}} [\| \mathcal{V}(v) - \mathcal{B}(s)) \|_2^2] \,, \tag{1}$$

where the brain encoder $\mathcal{B}$ learns the alignment between source brain space $\mathbf{B}$ and target image space $\mathbf{V}$.

Regression or Denoising. Regression using the Mean Squared Error (MSE, *i.e.*, L2) loss directly learns a deterministic mapping from brain data points to those of images [53]. However, one downside of this deterministic objective is that it can lead to overfitting, as the model may memorize specific instances rather than generalizing well to unseen data—especially considering the data scarcity in brain modalities. This issue becomes particularly problematic when dealing with noisy or incomplete data, as is often the case with brain signals, making the model sensitive to minor variations in the input. Denoising [16, 47, 48], on the other hand, has been shown to be more effective than regression, as introducing noise into training data acts as an implicit form of data augmentation and regularization [26, 52]. The denoising process encourages the model to focus on the underlying data manifold rather than memorizing specific instance values [16, 40, 47, 48]. Furthermore, visual signals suffer from heavy spatial redundancy [14]; recent work has shown that such redundancy also exists in visual features [6].

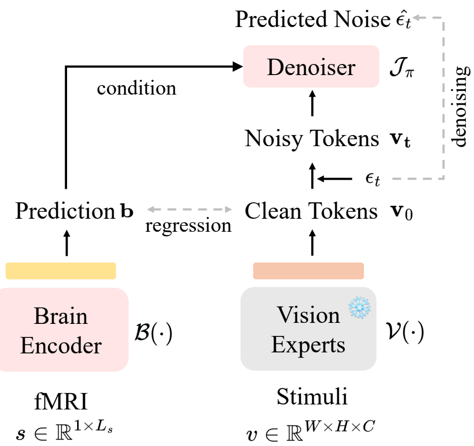

Figure 2. **Training Procedure with Denoiser.**

Therefore, we introduce a masked denoising optimization objective as shown in Fig. 2. The training procedure of the π-parameterized denoiser $\mathcal{J}_\pi$ follows a diffusion process:

$$\mathcal{L}_{\mathrm{D}} = \mathbb{E}_{\varepsilon,t} \left\| \mathcal{J}_\pi \left(\mathbf{v}_t; \mathbf{b}, t\right) - \varepsilon \right\|^2 . \tag{2}$$

Here, ε represents the ground truth noise sampled from the standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. The noise-corruption is defined as $\mathbf{v}_t := \sqrt{\bar{\alpha}_t}\mathbf{v} + \sqrt{1-\bar{\alpha}_t}\varepsilon$, where $\bar{\alpha}_t$ is a noise schedule [16, 33] at timestep t, and $\mathbf{v} = \mathcal{V}(v)$ maps input images v to its representation. The noise estimator $\mathcal{J}_\pi(\mathbf{v}_t; \mathbf{b}, t)$ predicts the noise level in $\mathbf{v}_t$ at timestep t, conditioned on brain predictions $\mathbf{b}$. The positional embedding [26] is added to the masked features to indicate the positions that need to be predicted, with the loss computed only on the unknown ones [14, 26]. The final loss as shown in Fig. 2 is the weighted sum of the regression loss and the denoising loss.

These features can be extracted from images on the fly during training or pre-calculated in advance to accelerate the process. During inference, the features predicted by the brain encoder, given brain signals as input, are fed into the remaining MLLM components (connector and LLM) for instruction-following tasks [29, 53], such as continuous dialogues, detailed descriptions, and complex reasoning.

4. MG-BrainDUB

To better evaluate the different choices of feature spaces in decoding fine-grained information, we propose a benchmark for detailed brain understanding. This involves curating high-quality evaluation datasets for two tasks: detailed captioning and salient question answering, each with corresponding metrics. We name this benchmark the Multi-Granularity Brain Descriptive Understanding Benchmark (MG-BrainDUB).

4.1. Descriptive Caption Understanding

Current brain captioning methods [12, 13, 49, 53] face two primary challenges. First, they rely on COCO [28] annotations, which provide only short captions with limited classes, lacking the granularity and vocabulary needed for detailed descriptions. Second, existing evaluation metrics are inadequate for assessing fine-grained captions. Traditional rule-based metrics, such as BLEU [36], CIDEr [51], and METEOR [4], compute n-gram matching scores between candidate and reference captions, making them highly sensitive to stylistic variations and leading to inconsistent evaluations. Model-based metrics, such as SPICE [3], suffer from outdated text encoders as backbones, while CLIP-based alternatives [15, 38] enforce a strict 77-token input limit, further restricting their effectiveness in evaluating detailed captions.

We tackle both challenges through two key strategies. For detailed caption annotation, we construct ground truth descriptions using state-of-the-art MLLMs [29], followed by manual error correction, missing element addition, and hallucination removal through expert human intervention. For evaluation metrics, in addition to leveraging CAPTURE [9], a metric specifically designed for long captions, we introduce a structured evaluation framework that assesses captions based on precision, recall, and F1 scores of three core visual elements: objects, attributes, and relations [9, 32]. To achieve this, we develop an automated pipeline with two key steps: visual element extraction and structured matching.

(a) Element Extraction. Given detailed captions, we first split them into separate sentences using NLTK toolkit [30] and extract key visual elements—objects, attributes, and their relations—using a T5-based factual parser [27, 39].

(b) Structured Matching. The matching process aligns extracted objects, attributes, and relations in three steps: exact matching (identifying precise word matches), synonym matching (matching words with similar meanings), and semantic matching (computing cosine similarity of word embeddings for any remaining unmatched elements).

Finally, we compute the precision, recall, and F1 scores for visual elements based on the matching results. These metrics capture semantic discrepancies: precision penalizes hallucinations and recall captures omissions, reducing confounds from pretrained generative model biases and offering a more accurate measure of genuine brain-based decoding.

4.2. Question-Answering Annotation

Previous studies have emphasized the multimodal evaluation for decoded *salient* objects, considering semantic selectivities in the higher visual cortex of the human brain [8, 18, 37]. Tiny objects in stimuli may not capture the subject's attention or relevant brain activities may not be effectively recorded during experiments [2, 53]. Thus, we further design evaluation focused on salient objects using annotated question-answering pairs. First, we extract salient objects using SAM [22] and input the cropped regions into the MLLM to identify objects, attributes, and relationships within each image using pipelines introduced in Sec. 4.1. We use SAM

Table 1. **Concept Localization.** We present brain grounding [53] results derived directly from brain signals, reflecting spatial information encoded in brain activities. Image-based results are included as references, comprising visual grounding from Shikra [7] using associated visual stimuli and grounding performance based on reconstructed images from state-of-the-art visual decoding methods [35, 41, 53, 54]. We highlight the best, second-best, and third-best performance averaged across all four subjects.

	Method	All		Salient		Salient Creatures		Salient Objects		Inconspicuous	
		acc@0.5	IoU	acc@0.5	IoU	acc@0.5	IoU	acc@0.5	IoU	acc@0.5	IoU
Vision	Shikra [7]	51.96	47.22	62.92	56.44	66.71	59.34	58.79	53.27	38.29	35.71
	BrainDiffuser [35]	17.49	19.34	27.18	27.46	38.71	34.63	14.62	19.66	5.39	9.20
	MindEye [41]	15.34	18.65	23.83	26.96	29.29	31.64	17.88	21.86	4.74	8.28
	DREAM [54]	16.21	18.65	26.51	27.35	34.43	33.85	17.88	20.28	3.35	7.78
	UMBRAE [53]	16.83	18.69	27.10	27.55	34.14	33.65	19.44	20.92	4.00	7.64
Brain	UMBRAE-S [53]	14.31	18.13	22.28	25.82	26.32	29.58	17.88	21.72	4.37	8.53
	UMBRAE [53]	18.03	20.63	28.22	29.32	36.93	35.61	18.74	22.47	5.32	9.79
	VINDEX-S	14.92	18.55	23.38	26.28	28.00	30.29	18.35	21.92	4.37	8.91
	VINDEX	18.10	20.89	28.50	29.78	37.00	36.20	19.25	22.78	5.11	9.80

instead of the original COCO location annotations or Referring Expression Comprehension [7] with COCO classes [28] to extract such information as pseudolabels, since COCO often overlooks salient categories outside its 80 common classes. For instance, an image may feature a large church, but only two small clocks on it are annotated with their spatial locations and linguistic descriptions. For each identified object, we design questions such as "which description best fits the <obj> in the image" and "what is the relationship between <obj1> and <obj2>' to evaluate the model's ability to recognize objects and their interactions within the image. Additionally, we include in-depth reasoning questions [29] that require step-by-step logical processing, based on GPT annotations [1, 43], to assess deeper reasoning capabilities.

To mitigate the MLLM's tendency to default to affirmative responses such as "Yes, I can see it" rather than genuinely recognizing an object, we follow [32] and introduce an equivalent number of question-answer pairs for objects not present in the image as hallucination components. We then ask the model to select the correct description from multiple choices, where one option is a correct caption, and two are incorrect. The attribute annotations of fine-grained objects serve as the answers, while erroneous linguistic annotations are manually modified to ensure they contradict the actual visual content. We refer to this as salient QA (SQA) for complex reasoning and report the response accuracy.

5. Experiments

In our experiments, we demonstrate multimodal interaction with fMRI using MLLMs to tackle four tasks: concept localization (Sec. 5.2), concise captioning (Sec. 5.3), descriptive captioning (Sec. 5.4), and complex reasoning (Sec. 5.5). While the generated multimodal explanations have been shown to enhance reconstruction performance [43, 49, 53], we prioritize brain perception over reconstruction.

5.1. Implementation Details

Architecture. For the mixture of vision encoders, we use pre-trained CLIP [38] and DINOv2 [34] as visual encoders. The target image features are extracted from both transformer encoders and then interleaved to spatially mix visual tokens before being subsequently processed by the connector and LLM [50]. The feature dimension D is 4,096 for 7B models and 5,120 for 13B models. For aggregated features, we utilize SigLIP [58] to extract visual features from multiple layers [57]. The connector first encapsulates multi-layer visual features and then maps the integrated visual features to the token space. For nested features, we use CLIP [38] to encode visual cues in a coarse-to-fine manner [6], where an image is represented as patch-based visual tokens. We create M sets of tokens, with the visual tokens at the coarser level derived directly from those at the finer level. Specifically, starting from the initial visual tokens, we sequentially apply 2×2 pooling with a stride of 2, resulting in 12×12, 6×6, and 3×3 visual tokens. Finally, we apply 3×3 pooling to obtain the most condensed single visual token. Details on encoders and MLLMs can be found in the supplementary.

Training Details. Our models are trained on a single A100 GPU. CLIP-224 [38] and DINO-224 [34] feature alignment with a batch size of 128 takes approximately 8 hours, CLIP-336 [38] with a batch size of 64 takes about 23 hours, while SigLIP-384 [58] with a batch size of 4 requires around 40 hours to converge. We use AdamW [31] as the optimizer with β_1=0.9, β_2=0.95, weight decay of 0.01, and apply the one-cycle strategy [46] with an initial learning rate of 3e-4 for the learning rate scheduler. Following visual decoding studies [41, 49, 54], we use the standard train and test splits for four subjects (`sub01`, `sub02`, `sub05`, `sub07`), each with 24,980 training samples and a shared 982 test samples. For evaluation, we report the average of the three repetitions of the same image in the test set, totaling 982 samples per subject. We implement the denoiser as an MLP with a default

Table 2. **Brain Captioning**. We evaluate short captions on BrainHub [53] and detailed captions on MG-BrainDUB. Models with '-S' refers to those with cross-subject support but are trained in a single-subject setting for comparison. MindEye2 results are based on a model adapted to one subject using a pre-trained model that was trained on the remaining seven subjects with the full dataset. The mark† denotes *zero-shot* methods, indicating that no external captions were used during training. 'Shikra-img' refers to image captioning results from Shikra [7] using corresponding visual stimuli as input, acting as an approximate performance upper bound. Colors denote best, second-best, and third-best performance. **Bold** and <u>underline</u> represent the best and second best results under identical protocols.

Method	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE	CIDEr	SPICE	CLIP-S	RefCLIP-S
					Concise Captioning (Sec. 5.3)					
Shikra-img [7]	82.38	69.90	58.63	49.66	35.60	65.49	161.43	27.62	80.60	85.92
SDRecon [49]	36.21	17.11	7.72	3.43	10.03	25.13	13.83	5.02	61.07	66.36
OneLLM [13]	47.04	26.97	15.49	9.51	13.55	35.05	22.99	6.26	54.80	61.28
BrainCap [12]	55.96	36.21	22.70	14.51	16.68	40.69	41.30	9.06	64.31	69.90
NeuroVLA [43]	57.19	37.17	23.78	15.85	18.60	36.67	49.51	12.39	65.49	-
MindEye2 [42]	54.82	38.60	26.49	18.16	17.54	43.77	55.70	10.97	67.54	73.73
MEVOX [55]†	58.56	40.36	28.09	20.11	19.20	44.47	54.37	11.05	64.23	70.36
UMBRAE-S†	57.63	38.02	25.00	16.76	18.41	42.15	51.93	11.83	66.44	72.12
UMBRAE [53]†	59.44	40.48	27.66	19.03	19.45	43.71	61.06	12.79	67.78	73.54
VINDEX-S†	57.99	39.00	26.04	17.83	18.56	42.27	53.96	11.92	66.94	72.64
VINDEX†	60.00	40.72	27.57	18.91	19.51	43.78	60.32	12.84	68.26	73.88
					Descriptive Captioning (Sec. 5.4)					
NeuroVLA [43]	38.91	24.02	15.24	12.41	18.44	27.83	42.58	18.41	56.16	-
VINDEX										
w/ SE	22.71	12.89	7.21	4.11	13.95	25.43	14.05	8.43	62.90	67.72
w/ ME	21.39	11.86	6.31	3.39	11.31	17.60	6.04	6.43	60.85	65.96
w/ AF	0.09	0.01	0.00	0.00	0.04	1.89	0.03	0.01	53.17	59.86
w/ NF_1	20.10	11.32	6.14	3.40	14.67	21.15	1.47	<u>9.85</u>	60.35	63.01
w/ NF_9	**50.77**	**33.25**	**21.17**	**13.20**	**18.26**	**41.72**	**46.50**	**11.45**	**65.40**	**71.20**
w/ NF_{36}	<u>38.87</u>	<u>23.87</u>	<u>14.39</u>	<u>8.80</u>	<u>14.72</u>	<u>33.40</u>	<u>31.07</u>	8.94	<u>64.02</u>	<u>69.80</u>
w/ NF_{144}	3.74	2.22	1.27	0.74	3.46	10.45	6.58	1.69	57.14	63.77

depth of 1, a latent dimension (width) of 1024, and a weight multiplier of 1.0 for the denoising loss (indicating the same contribution as the regression loss during training).

5.2. Concept Localization

This section evaluates the model's ability for concept localization through the brain grounding task [53]. Similar to visual referring expression comprehension, brain grounding uses the prompt "Find the coordinates of `<expr>`" and evaluates performance using accuracy and IoU metrics. The accuracy metric, `acc@m`, measures the percentage of correctly labeled instances with an IoU greater than the threshold m. We assess grounding results derived directly from brain signals, with image-based results provided for reference. These include visual grounding using ground truth stimuli and reconstructed images from state-of-the-art (SOTA) visual decoding methods [35, 41, 54] via Shikra [7]. We provide a comparison with SOTAs and our methods in Tab. 1.

The results demonstrate that our method VINDEX outperforms all baselines across all metrics in the four salient-based categories in BrainHub [53]. Notably, VINDEX surpasses both direct brain signal-based predictions from UMBRAE [53] and image-based grounding results, which use reconstructed images from brain signals as input [35, 41, 53, 54]. This emphasizes that directly extracting location information from the captured brain activities is more accurate than relying on reconstructed images. This observation suggests that reconstruction methods relying on image and text semantic encoding have inherent limitations in decoding spatial information. It also implies that incorporating grounding information in future models could help enhance the accuracy of object location prediction when reconstructing the visual stimuli from neural activations.

5.3. Concise Captioning

We evaluate short captions on BrainHub [53], demonstrating our method's performance on multimodal language tasks. Tab. 2 presents an evaluation of our brain captioning compared to state-of-the-art baselines, including three subject-specific methods SDRecon [49], BrainCap [12], OneLLM [13], and four cross-subject methods NeuroVLA [43], MindEye2 [42], MEVOX [55], and UMBRAE [53]. Our method, VINDEX, for concise captioning is built upon the same base MLLM Shikra [7] as in UMBRAE [53] for a fair comparison. MindEye2 is first pre-trained on seven subjects and then adapted to the remaining subject using all available data. NeuroVLA [43], UMBRAE [53], and VINDEX are trained on four subjects

Table 3. **Descriptive Captioning and Complex Reasoning**. We evaluate detailed descriptions and complex reasoning with salient question-answering on MG-BrainDub under different model settings. 'Space' refers to the feature space elaborated in Sec. 3.1, while 'size' and 'encoder' indicate the size of ground truth images and the encoder used for feature extraction. Colors denote best, second-best, and third-best performance. Models marked with $*$ correspond to those with the same settings as in Tab. 2.

Model Setting				CAPTURE	Object			Attribute			Relation			Runtime (s)	SQA
Space	Size	Encoder	MLLM		P	R	F1	P	R	F1	P	R	F1		Acc (%)
SE*	224	CLIP	LLaVA-1.5 7B	0.4356	57.22	49.76	51.88	37.90	36.08	35.73	45.25	40.70	42.32	14.05	75.50
SE	224	CLIP	LLaVA-1.5 13B	0.4733	61.81	52.91	55.60	40.68	41.21	39.38	49.28	45.00	46.52	8.70	77.92
SE	224	CLIP	LLaVA-1.6 7B	0.4684	51.02	58.78	53.51	36.01	54.01	41.80	41.00	45.63	42.75	10.91	76.80
SE	224	CLIP	LLaVA-1.6 13B	0.4645	50.44	57.83	52.81	35.89	54.02	41.72	40.42	45.44	42.38	24.18	78.69
ME*	224	CLIP, DINO	LLaVA-1.5 13B	0.3041	49.17	34.30	38.07	26.10	23.09	23.37	33.08	26.86	28.86	5.06	52.66
AF*	384	SigLIP	LLaVA-1.5 7B	0.0058	1.89	1.07	1.27	0.15	0.11	0.13	0.01	0.00	0.00	5.69	15.17
$NF_1{}^*$	224	CLIP	LLaVA-1.5 7B	0.4877	59.19	57.19	57.24	42.74	42.76	41.18	47.32	46.70	46.57	5.45	82.59
$NF_9{}^*$	224	CLIP	LLaVA-1.5 7B	0.5021	62.43	59.26	59.74	41.97	43.20	40.95	50.56	49.49	49.54	5.31	83.83
$NF_{36}{}^*$	224	CLIP	LLaVA-1.5 7B	0.4672	62.23	54.89	56.37	40.24	38.56	37.58	47.90	44.41	45.47	4.33	66.35
$NF_{144}{}^*$	224	CLIP	LLaVA-1.5 7B	0.1614	52.84	22.47	28.73	8.67	5.55	6.32	12.88	7.67	9.22	1.83	31.35
NF_1	224	CLIP	LLaVA-1.6 7B	0.4359	49.47	51.37	49.43	35.65	49.59	40.23	36.75	38.96	37.40	8.62	65.58
NF_1	336	CLIP	LLaVA-1.5 7B	0.4120	47.99	46.74	46.89	43.01	39.50	39.53	31.15	31.96	31.16	5.06	62.27

instead of eight. Caption results from Shikra [7] using associated visual stimuli for captioning act as an approximate upper bound. VINDEX achieves top performance across most metrics for concise brain captioning, even surpassing methods trained with more subject data [42] or external annotations [43]. The other brain-feature alignment methods UMBRAE [53] and MEVOX [55] ranked the second and the third. This highlights the effectiveness of the cross-subject brain feature alignment strategy, showcasing superior zero-shot performance without the need for external training data and generalization across subjects without requiring separate models or subject-specific parameters.

NeuroVLA, MindEye2, UMBRAE, and VINDEX generate fluent, complete, and information-rich sentences, highlighting the advantages of using LLMs. Notably, VINDEX and UMBRAE achieve superior performance in a zero-shot setting, without relying on additional annotations, outperforming models trained with real captions from COCO [28] or pseudo captions from MLLMs [29]. This demonstrates that directly aligning brain signals with image features offers a better way than using external visual, spatial, or linguistic annotations. It preserves more accurate semantic and spatial cues decoded from brain signals and avoids factual errors or stylistic inconsistencies from external annotations.

5.4. Descriptive Captioning

We present a quantitative comparison of descriptive captioning results for NeuroVLA [43] and our method with different feature space choices: single encoder (SE), mixture of encoders (ME), aggregated feature (AF), and nested features (NF_n) where the token size n is selected from $\{1, 9, 36, 144\}$ after downsampling. The latter Tab. 2 presents evaluations using regular caption metrics from [53]. Considering the limitations of traditional rule-based metrics (sensitive to writing style) and model-based metrics (unable to process long sequences), we evaluate descriptive captioning on MG-BrainDUB (Sec. 4.1). Results in Tab. 3 show that using NF_9 (with 9 visual tokens) as supervision for training brain encoders achieves the best performance on most metrics. Surprisingly, using dense feature aggregation (AF) instead yields the worst results, and models trained with vanilla CLIP features [38] (SE setting at first four rows) remain inferior, even with larger and more sophisticated MLLMs (scaling from v1.5 to v1.6, 7B to 13B).

Training with feature spaces from larger CLIP models (336 vs. 224) does not improve performance but significantly increases training time and computational costs. The brain encoder aligned with CLIP-ViT-L-224 features (corresponding to 16×16 tokens) generally transfers well to MLLMs designed with the CLIP-ViT-L-336 model (24×24 tokens) as the visual encoder, often yielding better results. AF performs the worst but with the largest token size (729 visual tokens for SigLIP-384 [58]). In contrast, aligning brain encoders with CLIP-224 requires only one-fifth of SigLIP-384's training time and one-third of CLIP-336's. Recall that NF downsamples the predicted CLIP features (the same features used in SE/ME) using pooling operations. Notably, 9-token alignment yields the best results, while 1-token alignment ranks second, suggesting that fewer tokens, though less expressive, align better with brain signals, which encode less information than real images. This also explains the inferior results for ME, where mixing brain-CLIP and brain-DINO features amplifies prediction imprecisions, leading to worse performance than using brain-CLIP features alone.

The performance drop with increasing feature magnitude suggests that, while richer image features benefit visual perception tasks [6, 50, 57], brain signals capture relatively less comprehensive visual elements. This may due to subjects' selective attention [8, 18, 37] or limitations in capturing brain activities [2, 53]. We expect additional features to enhance the perception of subtler elements, but they mainly introduce unnecessary hallucinations and increase training costs, as brain signals predominantly encode basic visual elements.

Results in Tab. 3 also indicate that traditional caption metrics [4, 36, 51] are not reliable when scaling to longer, more detailed captions. The rule-based metrics show that ME has higher scores than NF_1, even though NF_1 predicts more accurate objects, attributes, and relations. CLIP-based metrics [15] also show deficiencies, as they present high similarities for all generated captions, even for AF, which in most cases lead to nonsensical outputs, such as garbled text and blank spaces. The inference time does not increase as expected with the model complexity and token length; instead, it exhibits irregularities. $NF_9{}^*$ presents the best results across most metrics while maintaining low computation cost in terms of inference time and memory consumption.

5.5. Complex Reasoning

We evaluate the model's ability to perform complex reasoning on salient objects using Salient Question Answering (SQA) (Sec. 5.5), with accuracy reported in Tab. 3. Unlike previous sections focused on visual content, SQA involves in-depth reasoning that demands step-by-step logical processing to provide the right answer. This experiment highlights the model's ability to decode intricate characteristics of *salient* elements, meaning the prominent objects most likely to capture the subject's attention during brain activity capture [2], which have been shown to be better preserved in brain data [53]. From the table, $NF_9{}^*$ achieves the best SQA performance across image sizes, visual encoders, and MLLM settings, likely benefiting from a balance between rich visual features and brain data constraints, consistent with detailed description results. In contrast, ME and AF sometimes produce meaningless outputs due to improper feature-brain signal alignment, caused by the added complexity of features during training. They also suffer from hallucinations, generating non-existent objects and misrepresenting attributes and relationships, resulting in accuracy rates at or below random guessing.

6. Ablation Studies

Model-Agnostic Evaluation. VINDEX is a versatile, model-agnostic brain decoding framework that can be easily adapted to various subjects, image sizes, visual encoders, MLLM settings, input prompts, and downstream tasks. Due to the architectural consistency of LLaVA-based variants [6, 7, 29, 50, 57], our trained brain encoders can be directly integrated into different MLLMs without requiring additional modifications. Specifically, for model setups presented in Tab. 3, we only need to train four brain encoders using CLIP-224, CLIP-336, DINO-224, and SigLIP-384. The same brain encoder trained with CLIP-224 can be used across SE, ME, and NF (except for the last one with a 336 image size). This brain-CLIP features are then passed to the remaining MLLMs components, including connectors and LLMs, with task prompts supported by the base MLLM for multimodal interactions, such as interactive dialogue, brain captioning, complex reasoning, and concept localization. The brain encoder aligned with CLIP-224 features adapts well to LLaVA-1.6-7B/13B models, which are designed to use CLIP-336 as the visual encoder, leading to improved performance. This versatility enables our method to efficiently align brain signals with multiple MLLMs at minimal cost.

Table 4. **Ablation Study on Denoiser**. CAPTURE [9] is reported across architectural depth d, width w, and weight multiplier β.

Setup	S0	S1	S2	S3	S4	S5	S6	S7
β	-	1.0	1.0	1.0	1.0	0.5	1.5	2.0
d	-	1	2	3	1	3	3	3
w	-	512	1,024	1,024	1,024	1,024	1,024	1024
C	0.4179	<u>0.4308</u>	0.4257	0.4183	**0.4356**	0.4193	0.4197	0.4096

Denoising Optimization Objective. We conduct ablation studies on the denoiser in Tab. 4 to evaluate the effectiveness of the denoising objective over vanilla regression across different settings, focusing on architectural depth d, width w, and the weight multiplier β. Here, 'Sn' represents different configurations of specific depth, width, and weight multiplier; S0 refers to the base setup with only the regression loss, without the denoising loss. The results are reported as CAPTURE scores [9] (C) for detailed descriptions across each setup. Total training hours for 300 epochs across all settings is approximately eight hours, with no notable variations. We found that a simple one-depth MLP with 1,024 model channels yields the best results. Results show that training with the denoising loss setup incurs no extra cost but leads to improved performance and more stable training (further analysis in the supplementary material).

7. Conclusion

In this paper, we propose a zero-shot multimodal brain decoding method that explores representative feature spaces from the visual components within MLLMs. We elucidate different vision feature space choices and introduce a denoiser-based strategy to enhance brain alignment with pre-trained models, improving the accuracy and robustness of brain-encoded feature representations. Our method facilitates efficient and flexible decoding across multiple levels of granularity, eliminating the need for additional textual or spatial annotations during training. Furthermore, we present MG-BrainDUB, a benchmark that provides a more reliable and comprehensive evaluation for multigranular multimodal brain decoding. It comprises two core tasks—detailed description generation and salient question answering—along with a novel metric that prioritizes key visual elements including objects, attributes, and relationships, offering a robust framework for assessing fine-grained brain perception.

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